Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing (with Business Management)**

Unit ID	**2232**
Assessment	**Computer Networks 2.0**
Assessment Type	**Global Examination**
Date	**Wednesday 4th June 2025**
Time	**09.00AM – 12.00PM**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

a) Explain the primary purpose of the Open Systems Interconnection (OSI) seven-layer model in defining network architecture **and** outline its key importance in modern network communication, including its role in standardising protocols, enabling interoperability, simplifying troubleshooting, and facilitating modular network design. **4**

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 4 marks.

The OSI (Open Systems Interconnection) model is a conceptual framework that standardises the functions of a telecommunication or computing system into seven distinct layers (1 mark). Its primary purpose is to promote interoperability between different systems and vendors by defining clear roles and responsibilities for each layer (1 mark). It helps network professionals troubleshoot issues by providing a structured way to isolate and resolve problems layer by layer (1 mark). The model allows manufacturers to develop hardware and software that can communicate across platforms, as long as each follows OSI standards (1 mark). Additionally, it improves communication system design and teaching by offering a universal reference for understanding and explaining complex network functions (1 mark).

b) Identify and describe the primary functions of **TWO (2) specific** OSI layers (excluding Layer 4, the Transport Layer), and for each layer, provide examples of protocols that operate at that layer, explaining their role in network communication. **6**

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Marks

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

Any TWO (2) of:

- ***The Application Layer (Layer 7) provides services directly to the end user (1 mark), such as email, file transfer, and web browsing (1 mark). It enables applications to communicate over a network. Protocols that operate at this layer include HTTP, FTP, and SMTP (1 mark).***
- ***The Presentation Layer (Layer 6) ensures that data is presented in a format that the receiving application can understand. (1 mark) handling data formatting, encryption, decryption, and compression.(1 mark). conceptually, these types of services belong here: JPEG, MPEG, GIF (and other image/video formats) (1 mark for any)***
- ***The Session Layer (Layer 5) establishes, manage, and terminate communication sessions between applications on different devices (1 mark). Key functions include Session Establishment and Termination:, Dialogue Control: Synchronisation and Checkpointing: Protocols conceptually at this layer include NetBIOS: RPC, SIP (1 mark for any)***
- ***The Network Layer (Layer 3) determines how data is routed between devices on different networks (1 mark). It uses logical addressing and path determination (1 mark). A key protocol here is IP, which ensures packets are delivered to the correct destination (1 mark), and ICMP, used for error reporting and diagnostics (1 mark).***
- ***The Data Link Layer (Layer 2) manages the physical addressing of devices (MAC addresses) (1 mark) and ensures error detection in frames transferred over a local network (1 mark). Protocols at this layer include Ethernet and PPP (1 mark).***
- ***The Physical Layer (Layer 1) is responsible for the transmission of raw binary data over a physical medium such as cables or fibre optics (1 mark). Devices and standards here include RJ45, hubs, and various transmission media (1 mark). Protocols used in this layer include Ethernet, Wi-Fi, and Bluetooth (1 mark).***

Total 10 marks

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Marks

Question 2

a) Describe the primary roles and functional characteristics of the Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) in network communication, including their key differences in reliability, connection management, data transfer mechanisms, and typical use cases. **5**

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

TCP (Transmission Control Protocol) is a connection-oriented protocol (1 mark) that ensures reliable and ordered delivery of data using acknowledgements and error checking (1 mark). It is commonly used in applications requiring accuracy, such as web browsing, file transfers, and email (1 mark).

UDP (User Datagram Protocol) is a connectionless protocol (1 mark) that sends data without guarantees of delivery, sequencing, or error correction (1 mark). It offers faster transmission (1 mark) and is suitable for time-sensitive applications like VoIP and online gaming (1 mark).

Both protocols operate at the Transport Layer of the OSI model (1 mark) and support different needs—TCP prioritises reliability, while UDP prioritises speed and efficiency (1 mark).

b) Explain the primary purpose of the Hypertext Transfer Protocol (HTTP) in client-server communication and describe in detail how it facilitates interaction between clients (e.g., web browsers) and web servers, including key mechanisms such as request-response cycles, message formats, and statelessness. **5**

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

The HyperText Transfer Protocol (HTTP) is a standard application layer protocol that enables communication between clients (such as web browsers) and web servers (1 mark). It follows a request–response model, where the client requests data and the server respond with content or a status message (1 mark). HTTP is stateless (1 mark), treating each request independently without retaining session information (1 mark). It typically operates on port 80 (1 mark), making it widely accessible (1 mark). HTTP is essential for accessing web pages and interacting with web-based systems (1 mark).

Total 10 marks

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Marks

Question 3

a) Explain the primary purpose of subnetting in computer networks and outline specific ways in which it improves network efficiency, including examples of its impact on traffic management, address utilisation, and scalability. **5**

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

Subnetting is the process of dividing a large IP network into smaller, logical sub-networks (1 mark). It helps improve network organisation and traffic management by grouping related devices (1 mark). Subnetting reduces broadcast traffic and congestion, improving overall performance (1 mark). It also supports better access control (1 mark) and security through segmentation (1 mark). Additionally, it allows for more efficient use of IP addresses (1 mark) by matching subnet sizes to departmental needs (1 mark).

b) Explain the key features of star and mesh network topologies and discuss their relative advantages. **5**

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Marks

Mark Scheme:

1 mark per bullet point up to a maximum of 5 marks. Max 3 marks for each topology+ advantage

Star Network Topology

- ***All devices (nodes) are connected to a central hub, switch, or router.***
- ***Diagram***

***Relative Advantages*:**

- ***Simplified Troubleshooting: Because each device has a dedicated link to the central hub, isolating and diagnosing problems is much easier.***
- ***Easy Installation and Configuration: The setup is generally straightforward, as it involves connecting each device directly to the central hub.***
- ***Cost-effectiveness – fewer connections than mesh.***
- ***Centralised Control and Security: Network administrators can easily monitor and manage traffic***

Mesh Network Topology

- ***Devices are interconnected in a decentralised, web-like manner.***
- ***Some or all devices have direct connections to multiple other devices, creating multiple paths for data transmission.***
- ***Diagram (full or partial mesh)***

Relative Advantages:

- ***Exceptional Reliability and Redundancy: This is the strongest advantage of mesh topology. The existence of multiple paths means that the failure of a single link or node generally does not interrupt communication across the network. This is crucial for mission-critical applications.***

High Traffic Capacity: Mesh networks can handle very high volumes of traffic because multiple devices can transmit data simultaneously over different paths.

Total 10 marks

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Marks

Question 4

a) Compare the key features and performance differences between the IEEE 802.11a, 802.11b, 802.11g, 802.11n (Wi-Fi 4), and 802.11ac (Wi-Fi 5) wireless standards. Address the following aspects for each standard: **5**

- Frequency band(s) used
- Maximum theoretical data rate
- Channel width
- Modulation techniques
- Spatial stream support (MIMO)
- Backward compatibility
- Typical use cases and era of prevalence

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

802.11b operates at 2.4 GHz with speeds up to 11 Mbps (1 mark), offering wide compatibility but lower data rates (1 mark).

802.11a uses the 5 GHz band and supports speeds up to 54 Mbps (1 mark), providing better performance but shorter range due to higher frequency (1 mark).

802.11g also operates at 2.4 GHz and matches 802.11a in speed (up to 54 Mbps) (1 mark), but with improved backward compatibility and broader range than 802.11a (1 mark).

802.11n introduced MIMO (Multiple Input Multiple Output), operating on 2.4 and 5 GHz, and provides speeds up to 600 Mbps (1 mark), significantly improving bandwidth and signal strength (1 mark).

802.11ac, building on n, supports even higher speeds (up to 1 Gbps+) and only uses 5 GHz (1 mark), making it ideal for high-bandwidth tasks like HD video streaming and gaming (1 mark).

b) Recommend the most suitable IEEE 802.11 wireless standard for a busy open-plan office environment with high device density and frequent data transfers (e.g., video conferencing, cloud storage). Justify your choice by evaluating key performance factors such as frequency bands, channel capacity, interference resistance, data throughput, and support for modern features like MU-MIMO and OFDMA. **5**

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Marks

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

The most suitable wireless standard for a busy open-plan office is IEEE 802.11ac (or 802.11ax, if allowed) (1 mark), as it provides high throughput (1 mark), greater channel width (1 mark), and operates on the 5 GHz band, reducing interference from other devices (1 mark).

It supports speeds of up to 1 Gbps or more (1 mark), making it ideal for high-density environments with many users (1 mark).

802.11ac also includes features such as beamforming (1 mark) and multi-user MIMO (1 mark), which enhance signal quality and maintain stable connections across the open-plan area (1 mark).

The 5 GHz frequency offers less congestion and faster data rates than 2.4 GHz (1 mark), especially important in environments with streaming, VoIP, or cloud-based tools (1 mark).

Although the range is slightly shorter compared to 2.4 GHz (1 mark), the higher performance and interference resistance make 802.11ac a better choice for busy workspaces (1 mark).

Total 10 marks

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Marks

Question 5

a) Explain the primary purpose of a firewall in network security and detail its operational mechanisms for protecting a network. Address the following: **5**

- Core objectives (e.g., traffic filtering, access control, threat mitigation)
- Key types of firewalls (e.g., packet-filtering, stateful inspection, application-level gateway)
- Criteria used to filter traffic (e.g., IP addresses, ports, protocols, application layer data)
- Role in preventing unauthorised access and common cyber threats (e.g., malware, brute-force attacks)

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

A firewall is a security device or software application that monitors and controls incoming and outgoing network traffic (1 mark) based on predetermined rules (1 mark). It acts as a barrier between a trusted internal network and untrusted external networks (1 mark), such as the internet (1 mark). By inspecting data packets, it allows or blocks traffic depending on factors like IP addresses, port numbers, or protocols (1 mark). This helps prevent unauthorised access (1 mark), reduces the risk of attacks such as malware or hacking (1 mark), and enforces organisational security policies (1 mark). Firewalls may be implemented in hardware, software, or both (1 mark), and are typically placed at the network perimeter to protect the internal infrastructure (1 mark).

b) Describe how Virtual Local Area Networks (VLANs) **and** Access Control Lists (ACLs) contribute to internal network security. Specifically address: **5**

- The role of VLANs in segmenting network traffic, isolating sensitive data, and limiting lateral movement of threats.
- The function of ACLs in enforcing granular traffic filtering based on criteria like IP addresses, ports, protocols, and traffic direction.
- How the combination of VLANs and ACLs strengthens security by reducing attack surfaces, controlling inter-VLAN communication, and preventing unauthorised access between network segments.

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Marks

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

VLANs (Virtual Local Area Networks) enhance internal security by logically segmenting a network into isolated groups (1 mark), limiting broadcast traffic (1 mark) and restricting communication between departments unless explicitly allowed (1 mark). This minimises the risk of internal threats (1 mark) spreading across the entire network and enforces access boundaries (1 mark). Access Control Lists (ACLs) are used to define rules on routers or switches that permit or deny traffic based on criteria such as IP address, protocol, or port number (1 mark). ACLs allow administrators to control which users or devices can access specific resources within or across VLANs (1 mark), helping to prevent unauthorised access and maintain security policy enforcement (1 mark).

Total 10 marks

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Marks

Question 6

a) Describe the core principle of network segmentation in business networks **and** explain how it enhances both security and performance. Address the following: **4**

- Fundamental concept of dividing a network into logical segments (e.g., VLANs, subnets).
- Specific security benefits (e.g., limiting lateral movement of threats, isolating sensitive data, reducing attack surfaces).
- Performance improvements (e.g., traffic optimisation, reduced broadcast domains, efficient resource allocation).
- Practical examples of segmentation strategies (e.g., separating guest networks, isolating IoT devices, segmenting departments).

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 4 marks.

Network segmentation is the practice of dividing a larger network into smaller, isolated segments or subnets, often using VLANs or firewalls (1 mark). This limits the spread of malicious traffic (1 mark) and allows security policies to be applied more precisely to specific areas of the network (1 mark). Segmentation reduces congestion by containing broadcast traffic within each segment (1 mark), which improves overall network performance (1 mark). It also simplifies monitoring and management (1 mark), allowing IT teams to isolate issues and respond more quickly to faults (1 mark). In a business environment, segmentation supports secure access control (1 mark) and protects sensitive data by limiting unnecessary inter-departmental communication (1 mark).

b) Describe how each of the following commands or protocol is used in network fault diagnosis **and** how each assist in resolving network issues. **6**

- Ping
- Traceroute
- SNMP

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Marks

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 6 marks.

- ***Ping is used to test basic network connectivity by sending ICMP echo requests and measuring response times (1 mark), helping to determine if a device is reachable (1 mark).***
- ***Traceroute maps the path data takes to a destination by identifying each router along route (1 mark), allowing administrators to locate points of delay or failure (1 mark).***
- ***SNMP (Simple Network Management Protocol) collects performance and status information from network devices (1 mark), enabling real-time monitoring and alerting for faults or abnormal behaviour (1 mark).***

Total 10 marks

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Marks

Question 7

a) Describe the primary functions of the Dynamic Host Configuration Protocol (DHCP) and the Domain Name System (DNS) in a network environment. For each, address: **5**

- Core purpose and role in network operation
- Technical mechanisms or processes they employ
- How they enable connectivity and usability for end-users
- Interactions with other network components (e.g., routers, switches, client devices)

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, gateways, and DNS server details to devices on a network (1 mark), reducing manual configuration and ensuring address consistency (1 mark). DHCP simplifies device management (1 mark) and supports mobile and temporary connections (1 mark) by dynamically allocating network settings (1 mark). DNS (Domain Name System) translates human-readable domain names (e.g. www.example.com) into IP addresses (1 mark) that computers use to identify each other on a network (1 mark). This allows users to access websites and services without needing to remember numerical IP addresses (1 mark). Together, DHCP and DNS ensure smooth connectivity and efficient communication across the network (1 mark).

b) Explain how server software supports essential network services, including file sharing, print services, and centralised management. For each service, address: **5**

- Core functionality provided by server software
- Technical protocols or mechanisms used (e.g., SMB, NFS, CUPS, LDAP)
- How server software enables secure, efficient delivery of the service
- Role in facilitating centralised control and management of network resources

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Marks

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

Server software enables the sharing of network resources (1 mark), such as files and printers, by receiving requests from client machines and delivering the appropriate service if user permissions allow (1 mark). It handles file sharing by offering private and shared storage areas with configurable access rights (1 mark). Print services are supported by enabling multiple users or departments to access shared network printers, reducing the need for individual devices (1 mark). The software also manages centralised user permissions (1 mark), allowing administrators to assign, modify, or revoke access to resources through groups or domains (1 mark). Additionally, server software integrates all these services for simultaneous access by multiple users (1 mark), enhancing efficiency and simplifying network administration (1 mark).

Total 10 marks

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Marks

Question 8

a) Explain the key benefits and limitations of using Voice over Internet Protocol (VoIP) in a modern office network. Address the following for each category: **5**

- Benefits
  - Cost-efficiency (e.g., reduced infrastructure, long-distance savings)
  - Functionality (e.g., unified communications, scalability, advanced features like video conferencing)
  - Integration capabilities (e.g., with CRM systems, cloud services)
  - Flexibility for remote work/hybrid environments
- Limitations
  - Dependence on network infrastructure (e.g., bandwidth requirements, latency/quality issues)
  - Security vulnerabilities (e.g., eavesdropping, denial-of-service attacks)
  - Reliance on power/Internet connectivity
  - Potential compatibility challenges with legacy systems
- Practical considerations for balancing benefits and limitations in office deployments (e.g., QoS implementations, security protocols, redundancy planning)

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

Voice over IP (VoIP) allows voice communication over IP networks (1 mark), reducing costs by using existing data infrastructure instead of separate telephone lines (1 mark). It supports useful features such as voicemail, call forwarding, and conferencing, many of which are free or low-cost compared to traditional systems (1 mark). VoIP is scalable and flexible (1 mark), making it ideal for growing businesses and hybrid work environments (1 mark). However, it is susceptible to quality issues such as latency, jitter, and packet loss, (1 mark) especially if network bandwidth is limited or poorly managed (1 mark). It also shares the same security risks as general internet traffic (1 mark), including threats like eavesdropping, hacking, and denial-of-service attacks (1 mark), requiring strong encryption and secure configurations (1 mark).

b) Describe how a video conferencing system can be integrated into an existing LAN setup. **5**

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Marks

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

A video conferencing system can be integrated into an existing LAN by connecting user devices such as computers or IP phones equipped with webcams, microphones, and speakers to the local network (1 mark). Software platforms like Skype or Microsoft Teams are installed and configured to handle audio and video traffic (1 mark). The system uses existing IP infrastructure to transmit real-time media via protocols like RTP or SIP (1 mark), ensuring compatibility with network services such as DHCP and DNS (1 mark). Adequate bandwidth must be ensured, and routers or firewalls should allow traffic on relevant ports to support stable video calls (1 mark). Integration also requires configuring endpoint devices to communicate with internal servers or cloud-based services (1 mark), supporting one-to-one and group meetings securely and reliably (1 mark).

Total 10 marks

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Marks

Question 9

a) What is a Virtual Private Network (VPN), and how does it secure remote communication for employees? Address the following: **5**

1. Core Definition
2. Security Mechanisms
3. Protection of Remote Communication
4. Role in Remote Work Security
5. Practical Components in Employee Use Cases

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

A Virtual Private Network (VPN) is a secure communication method that allows users to transmit data across shared or public networks (1 mark) as if directly connected to a private network (1 mark). It uses encryption and tunnelling protocols to protect data (1 mark), giving the appearance of a dedicated private connection (1 mark). VPNs support remote access (1 mark), enabling employees to connect to internal company systems from external locations (1 mark), such as their homes or while travelling (1 mark). The data is transmitted through secure VPN tunnels, typically using protocols like PPTP, L2TP/IPsec, or OpenVPN, ensuring both confidentiality and integrity (1 mark). This allows staff to work safely over the internet while maintaining the security of corporate resources and communications (1 mark).

b) Describe a specific corporate scenario where implementing Virtual Private Network (VPN) access control policies would significantly enhance security. In your description, include the following details: **5**

1. The Business Context
2. The Security Threats
3. The VPN Access Control Policies
4. The Security Improvements

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Marks

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

In a business where employees frequently access the corporate network remotely, implementing VPN access control policies ensures that only authorised users can connect to sensitive systems (1 mark). For example, access could be limited by device type, time of day, user role, or geographical location, (1 mark) preventing unauthorised or risky connections (1 mark). A policy might allow full access to internal databases for finance staff, while restricting contractors to a shared folder only (1 mark). These controls help maintain confidentiality and data integrity by enforcing strict rules on who can access what (1 mark), and under what conditions (1 mark). They also reduce the attack surface by blocking unnecessary VPN usage and detecting unusual access patterns early (1 mark).

Total 10 marks

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Marks

Question 10

a) Propose a comprehensive network monitoring strategy that leverages Simple Network Management Protocol (SNMP) and log analysis tools. In your proposal, address the following aspects: **5**

- Objectives and Scope
- SNMP Implementation
- Log Analysis
- Alerting and Response
- Tools and Technologies
- Scalability and Maintenance

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

An effective network monitoring strategy uses tools such as SNMP (Simple Network Management Protocol) to collect data from network devices like switches, routers, and servers, enabling real-time monitoring of performance and availability (1 mark). SNMP agents report key metrics (e.g. CPU load, bandwidth usage) to a central manager, allowing early detection of faults or unusual activity (1 mark). Log analysis involves collecting system and event logs from devices, which can be reviewed manually or automatically to identify errors, failed logins, or security events (1 mark). Combining these methods provides a holistic view of the network, supports trend analysis, and ensures quicker response to incidents (1 mark). Using tools such as protocol analysers or traffic monitors also helps establish performance baselines and diagnose issues like packet loss or congestion (1 mark).

b) Explain, with specific examples, how maintaining up-to-date network documentation, including network diagrams and change logs, supports network troubleshooting and management processes. In your explanation, address the following aspects: **5**

- Role in Troubleshooting
- Benefits for Network Management
- Impact on Efficiency

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Marks

Mark Scheme:

Any suitable alternative phrasing or technically accurate response may be credited, up to a maximum of 5 marks.

Maintaining up-to-date network documentation, including diagrams and change logs, enables faster and more accurate troubleshooting by providing a clear record of network layout, device configurations, and changes over time (1 mark). Accurate network diagrams help IT staff quickly locate faults, understand interconnections, and assess impact areas during outages or upgrades (1 mark). Change logs record modifications such as configuration updates or hardware replacements, reducing the risk of errors and allowing rollback if problems arise (1 mark). This documentation also supports better planning for expansion or maintenance tasks, ensuring compatibility and consistency (1 mark). Overall, it promotes efficiency, reduces downtime, and serves as a valuable reference for new or external support personnel (1 mark).

Total 10 marks

End of paper

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Learning Outcomes Matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO1	Yes
2	LO1	Yes
3	LO1, LO4	Yes
4	LO1, LO2, LO4	Yes
5	LO3, LO4	Yes
6	LO3, LO5	Yes
7	LO1, LO5	Yes
8	LO6	Yes
9	LO3, LO5	Yes
10	LO5	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand network and communication protocols	Demonstrates little to no understanding of the concept of network.	Basic understanding with significant gaps.	Demonstrates a basic understanding of the concept of an engineering approach to software development.	Shows a good understanding of the concept of computer network.	Demonstrates an excellent and thorough understanding of the concept of computer network.
Understand the principles of common network topologies and architectures	Demonstrates little to no understanding of network topologies and architecture.	Shows a basic but limited understanding of network topologies and architecture.	Demonstrates a basic understanding of network topologies and architecture.	Shows a good understanding of network topologies and architecture.	Demonstrates an excellent and thorough understanding of network topologies and architecture.
Understand the application of network security measures	Demonstrates little to no skills of network security and its tools.	Shows a basic but limited skill of network security and its tools.	Demonstrates a basic skill of network security and its tools.	Shows a good skill of network security and its tools.	Demonstrates an excellent skill through network security and its tools.
Be able to select and configure the hardware components of a computer network to meet the requirements	Shows little to no understanding of network hardware components.	Basic understanding of network hardware components.	Adequate understanding of network hardware components.	Good understanding of network hardware components.	Excellent and comprehensive understanding of network hardware components.

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of a precise specification.					
Be able to design and install network and server operating systems to meet the requirements of a precise specification	Show little or no understanding of network software.	Basic understanding of network software.	Adequate understanding of network software.	Good understanding of network software.	Excellent understanding of network software.
Be able to define and configure internet telephony and communication systems	Show little or no understanding of internet telephony and communication systems	Basic understanding of internet telephony and communication systems	Adequate understanding of internet telephony and communication systems	Good understanding of internet telephony and communication systems	Excellent understanding of internet telephony and communication systems

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